Digital Image Transmission over Multipath Channels

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Abstract

A block-DCT image is quantized into 8-bit coefficients, transmitted as antipodal binary symbols through pulse-shaped multipath channels with additive white Gaussian noise (AWGN), and reconstructed following matched filtering, symbol-spaced Zero-Forcing (ZF) or Minimum Mean-Square Error (MMSE) equalization, and hard decisions. Half-Sine and square-root raised-cosine (SRRC) signaling are evaluated using waveform spectra, eye diagrams, bit-error rate (BER), reconstructed-image peak signal-to-noise ratio (PSNR), and alternate impulse responses. All numerical results are generated by the accompanying executed Jupyter notebook.

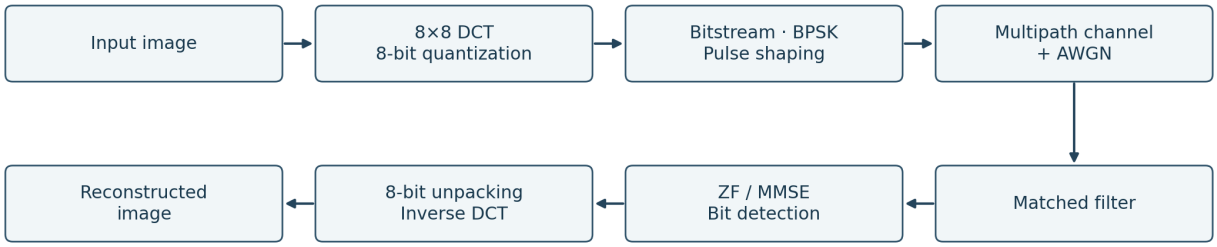


Figure 1. End-to-end image-transmission architecture.

Experiment configuration

Parameter	Configuration
Source image	64 × 128 pixels (grayscale); 65,536 transmitted payload bits
Source coding	8 × 8 orthonormal DCT; global range mapped to 256 levels; MSB-first bit packing
Modulation	Antipodal binary signaling (−1, +1); 32 samples per symbol
Pulse shaping	Unit-energy Half-Sine; SRRC (β = 0.5; K = 2 and 6)
Primary channel	Four symbol-spaced paths: [1, 0.5, 0.75, −2/7]
Receiver	Matched filter; raw decisions, FFT-based ZF, FFT-based MMSE
Evaluation	AWGN σ = 0.05–1.1; BER, received-image PSNR, matched-sample SNR

Quantizer minimum/maximum coefficients and image dimensions are assumed to be known without error at the receiver. The notebook models the communication channel rather than an autonomous image-file protocol.

1. Image source coding and symbol mapping

The 8 × 8 image blocks undergo an orthonormal two-dimensional discrete cosine transform (DCT). For each image, the global DCT coefficient interval [c_min, c_max] is uniformly mapped to integer values 0–255. The flattened coefficient bytes are serialized MSB-first to a 65,536-bit payload. Hard-detected bytes are reassembled, rescaled to the original coefficient interval, and inverted by IDCT. No entropy or error-control coding is applied.

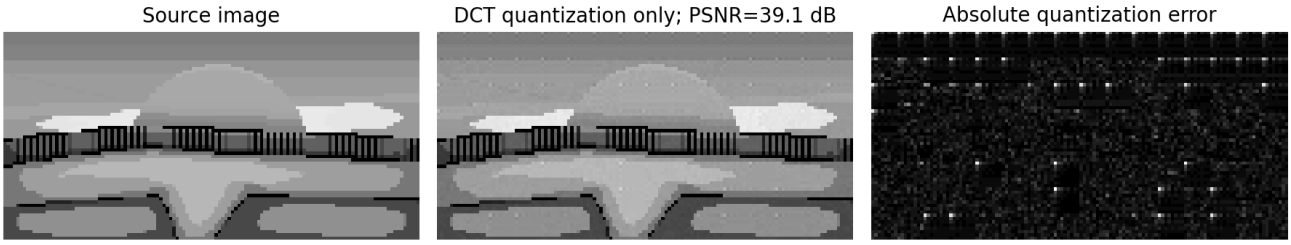


Figure 2. Original image, decoded image with DCT quantization alone, and the absolute quantization error.

Without any transmission errors, the quantized reconstruction reaches **39.09 dB** PSNR for the supplied source image. This is the image-quality ceiling of this fixed quantizer; a zero-BER link does not imply a pixel-perfect image.

2. Pulse design

The Half-Sine pulse is sin(πt/T) for 0 ≤ t < T, zero elsewhere. SRRC pulse coefficients are sampled on [−KT, KT] and normalized to unit discrete energy, with analytic limiting values at the removable singularities. In an ideal channel, the pulse/matched-filter cascade of SRRC approximates a raised-cosine Nyquist response. Its finite truncation leaves residual intersymbol interference (ISI).

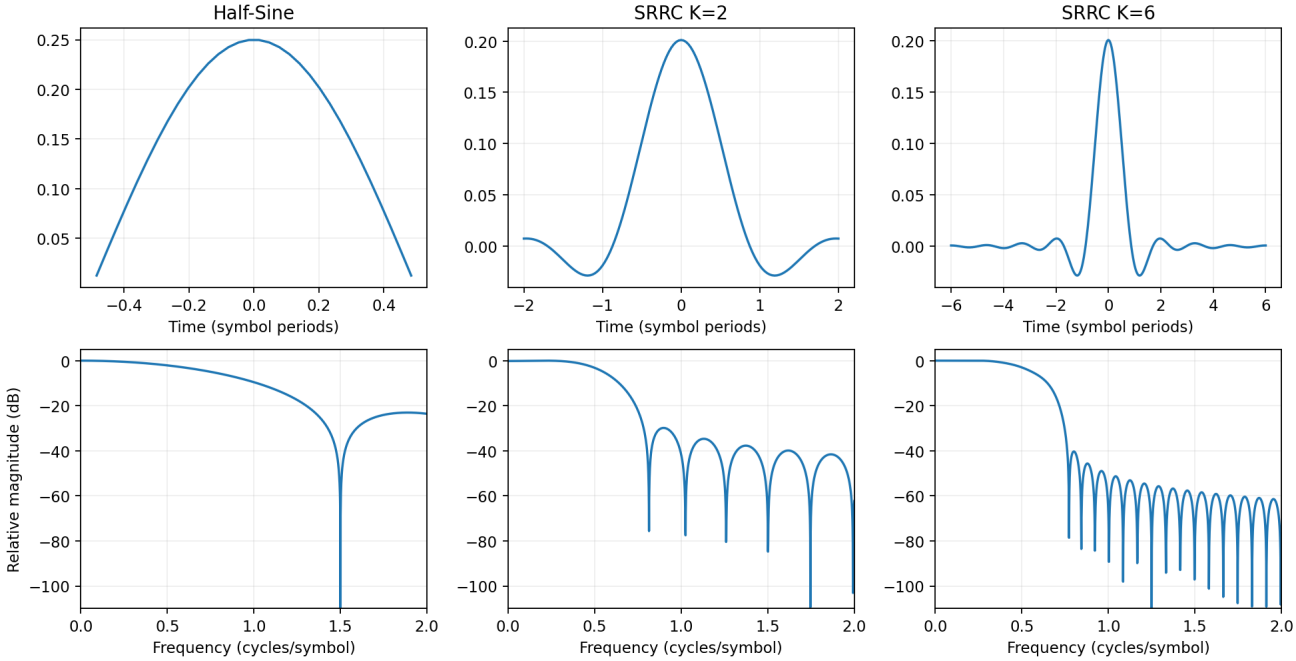


Figure 3. Energy-normalized Half-Sine and SRRC pulses, K = 2 and K = 6, with Fourier magnitude responses.

3. Transmit spectra and channel frequency response

The binary symbols are converted into an impulse train spaced by 32 samples and convolved with the chosen pulse. Welch power spectral density estimates capture the modulation spectrum. A longer truncated SRRC pulse produces more selective out-of-band behavior; its ideal nominal excess bandwidth is determined by β, not simply by its span K.

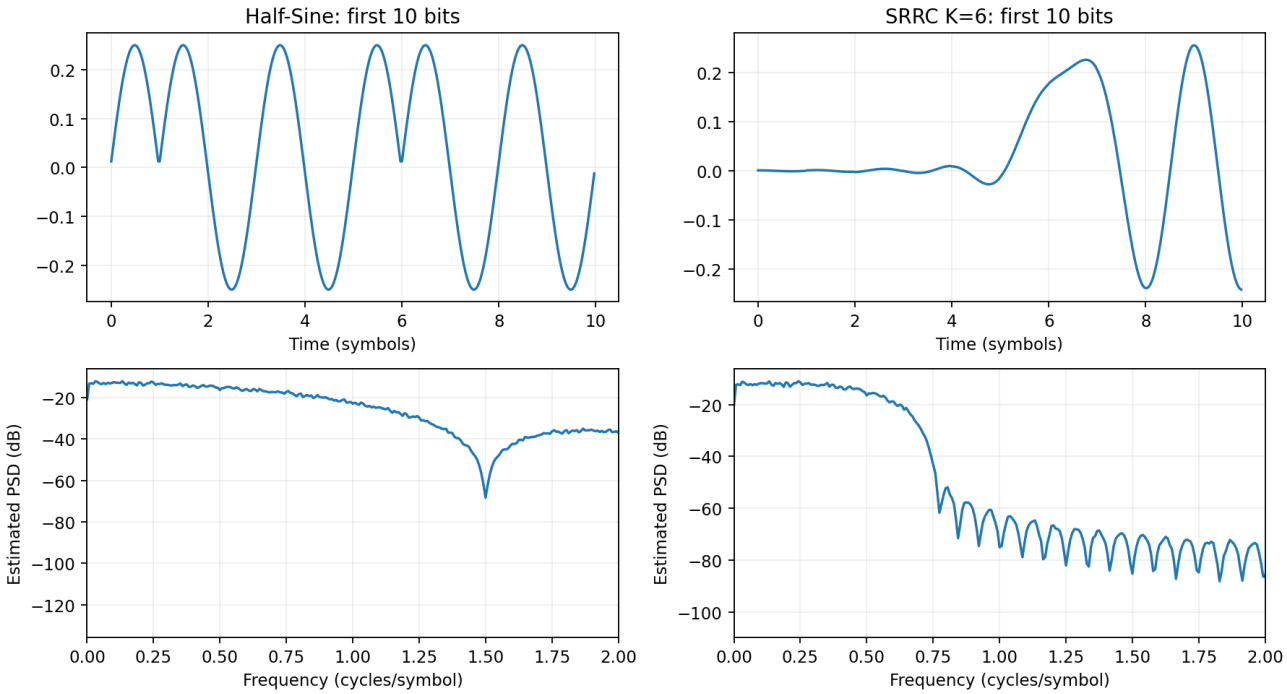


Figure 4. Example transmitted waveforms and empirical PSD estimates using the same bit sequence.

The principal multipath channel has symbol-spaced taps h = [1, 1/2, 3/4, −2/7]. These correspond to impulses at delays 0, T, 2T and 3T. Channel distortion follows from frequency-selective amplification and attenuation; severe spectral dips can make direct inversion ill-conditioned.

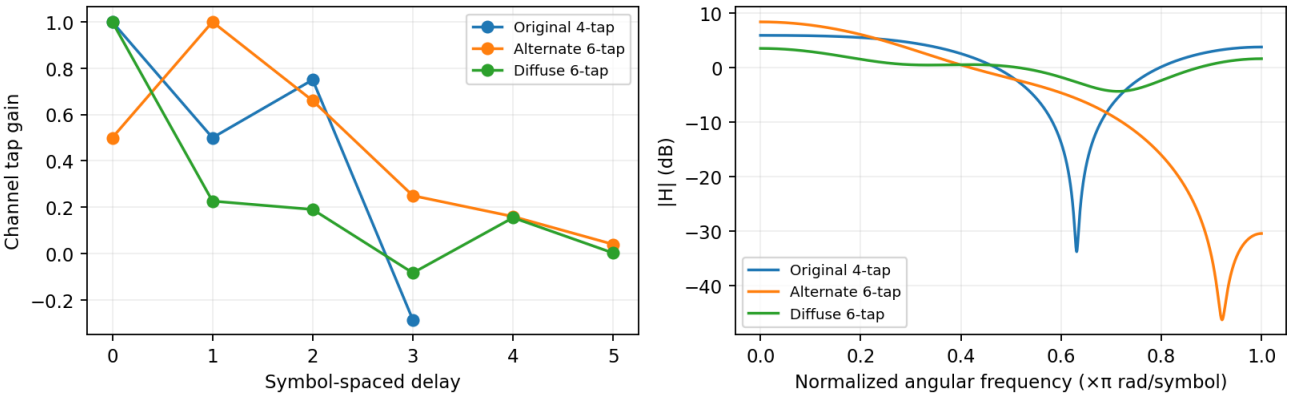


Figure 5. Impulse responses and frequency-selective magnitude responses of the principal and two alternate channels.

4. Matched filter, timing, and eye diagrams

The channel output is r[n] = (a ↑ 32 ∗ g_T ∗ h)[n] + v[n], with sample-wise zero-mean Gaussian noise v of standard deviation σ. For real-valued pulses, the matched-filter impulse response is the time reversal g_R[n] = g_T[L−1−n]. Decision samples are extracted at the zero-delay peak of g_T ∗ g_R and then every 32 samples. The matched filter optimizes detection SNR for an isolated known pulse in AWGN; it does not cancel channel-induced ISI.

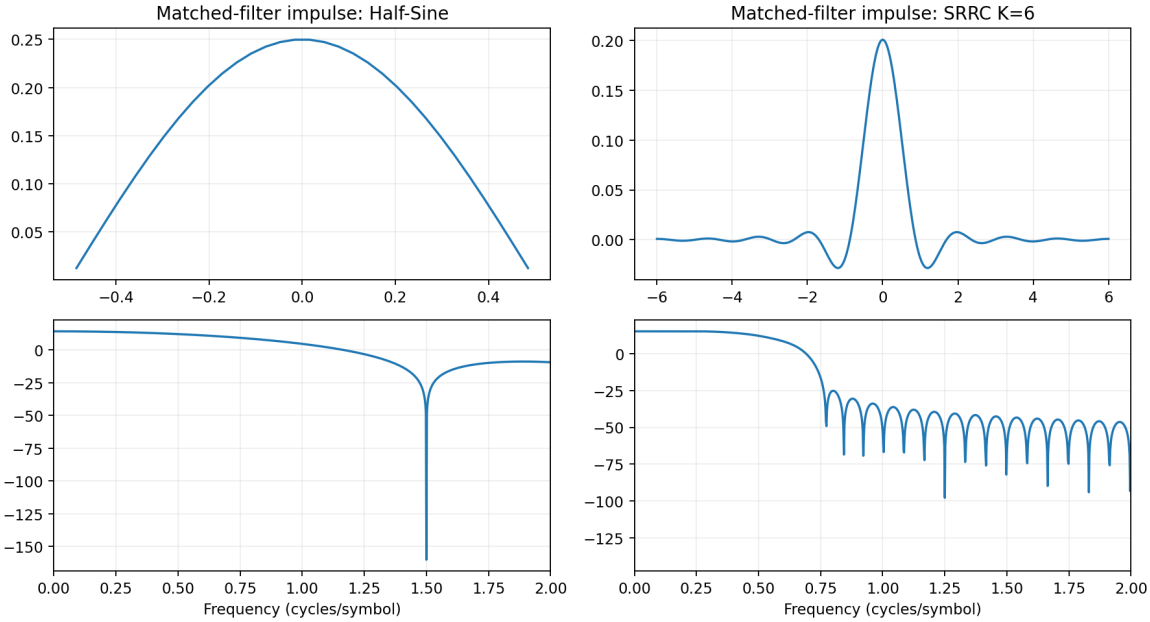


Figure 6. Matched-filter time- and frequency-domain response for both pulse families.

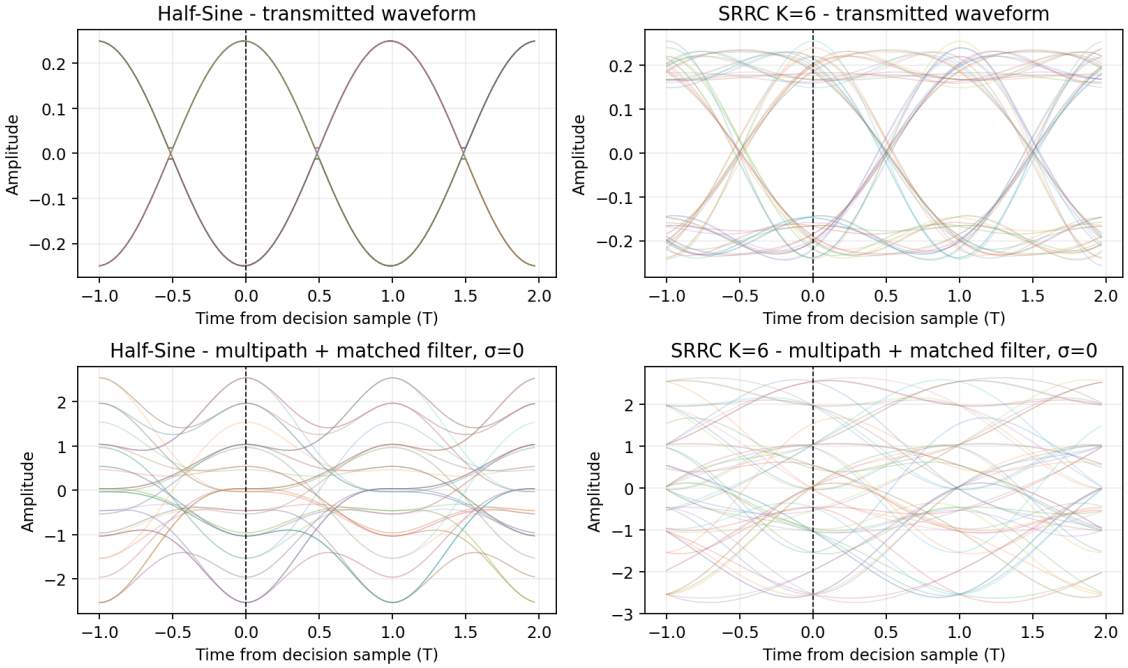


Figure 7. Transmitter and noiseless multipath/matched-filter eye diagrams; the dashed line marks the decision phase.

5. Noise sensitivity and equalizer design

Noisy eye diagrams were computed for σ = 0.1, 0.5 and 0.9. Larger noise obscures the decision opening and increases decision uncertainty. The receiver models the entire transmit-pulse/channel/matched-filter cascade as a symbol-spaced effective response G(e^{jω}); this step is necessary to avoid treating the raw multipath taps as the complete sampled channel.

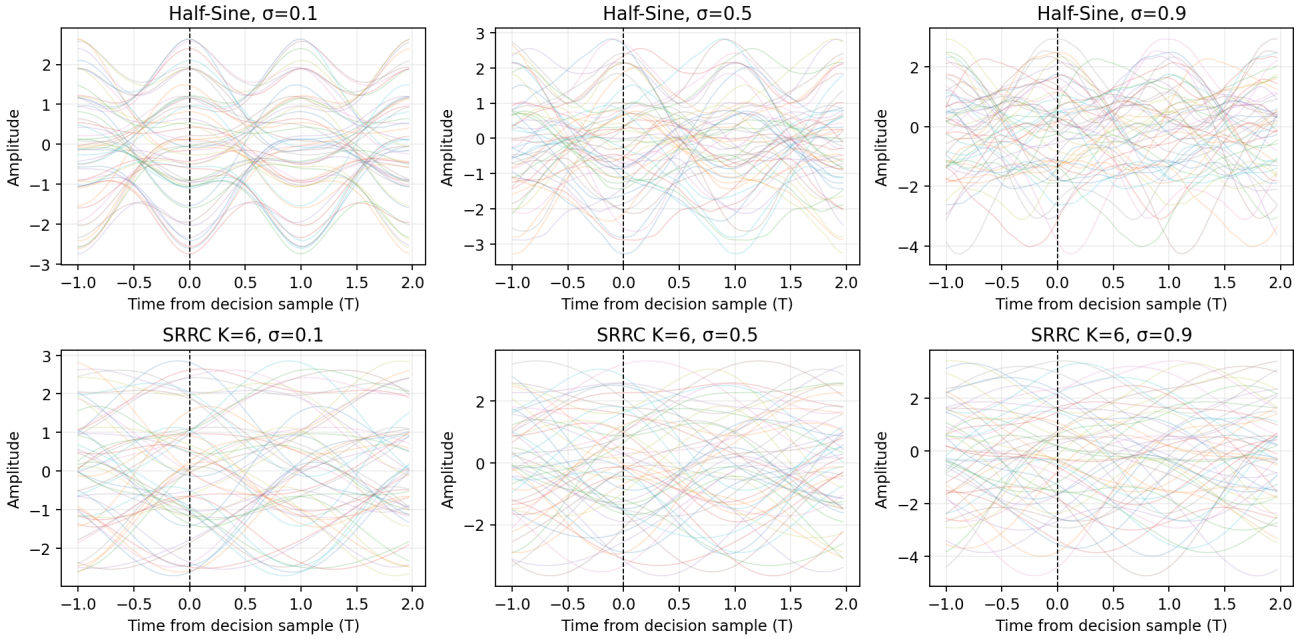


Figure 8. Received matched-filter eyes for increasing additive Gaussian noise.

The implemented approximate inverse filters are Q_ZF = G* / max(|G|², ε) and Q_MMSE = G* / (|G|² + σ²), for unit-variance binary symbols and an approximately white post-matched-filter noise model. The small ZF denominator floor is numerical protection, not evidence that an exact stable ZF inverse exists. MMSE is a regularized solution and may retain residual ISI.

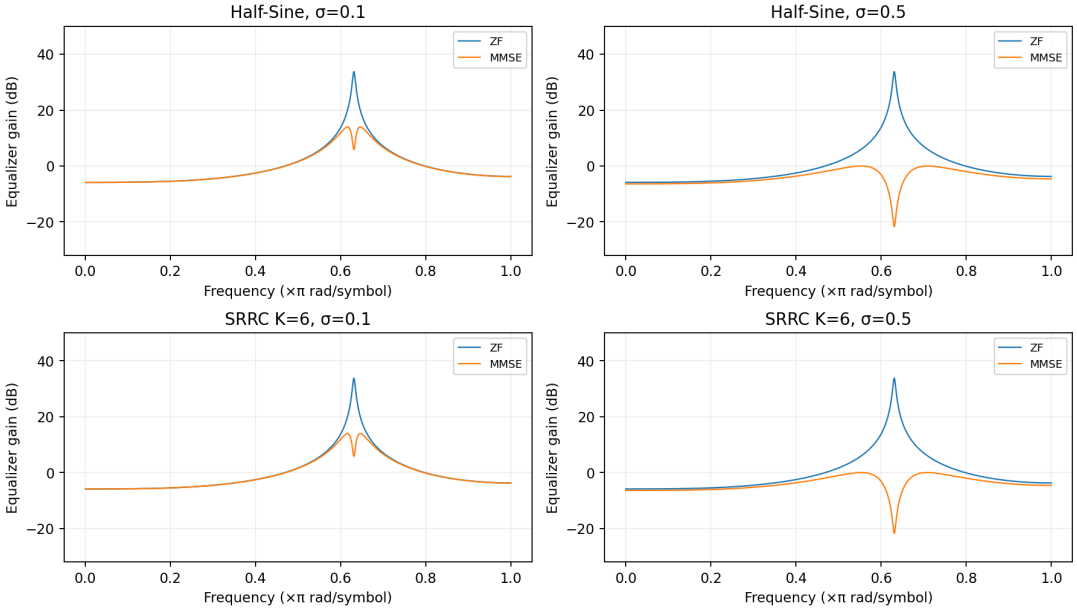


Figure 9. Symbol-rate ZF and MMSE frequency responses for two noise levels and both pulse shapes.

6. Equalizer waveform diagnostics and BER behavior

For eye-diagram diagnostics, sample-rate equalization of the multipath transfer function is additionally shown. These sample-rate eye displays are distinct from the symbol-spaced effective-response equalizers used for all measured BER and PSNR results. Their purposes must not be conflated.

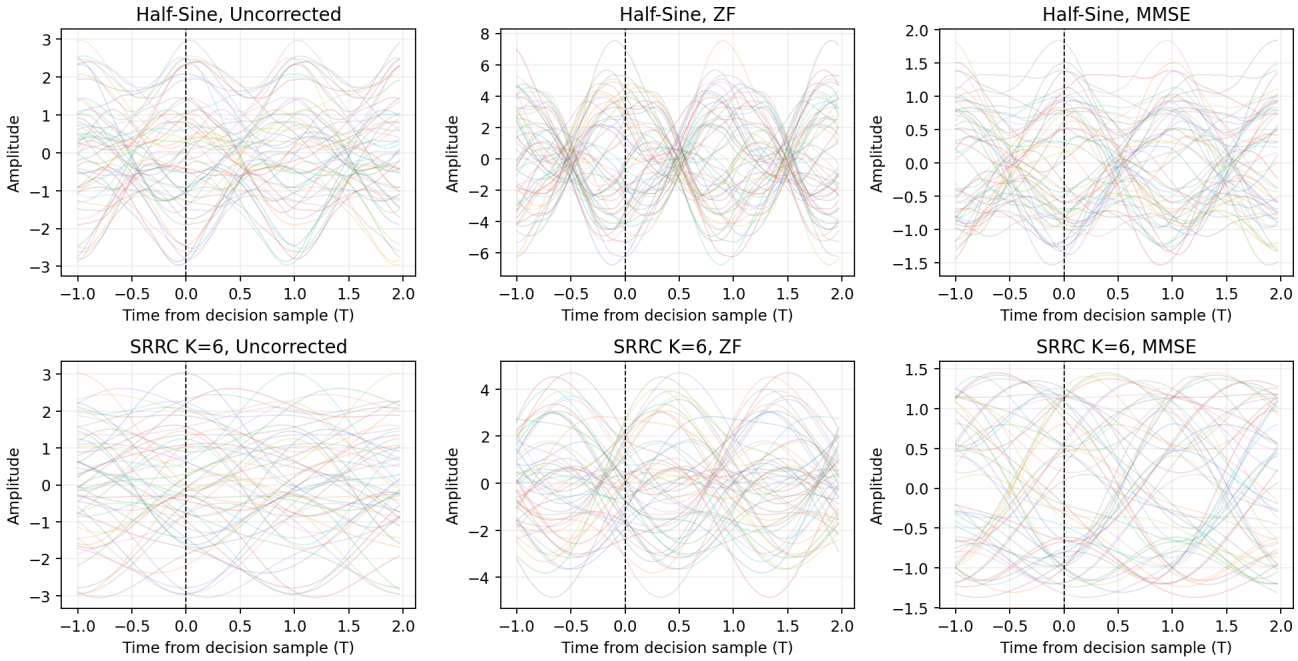


Figure 10. Waveform-rate eye-diagram diagnostics with uncorrected, ZF, and MMSE filtering (σ = 0.5).

For the original 65,536-bit image, BER is measured by a direct bitwise comparison. Although the data block is finite and FFT equalization is circular, the full payload is counted, including any small block-boundary contribution. The numerical results show strong ZF noise amplification under this channel at moderate and high noise; MMSE significantly limits the resulting BER.

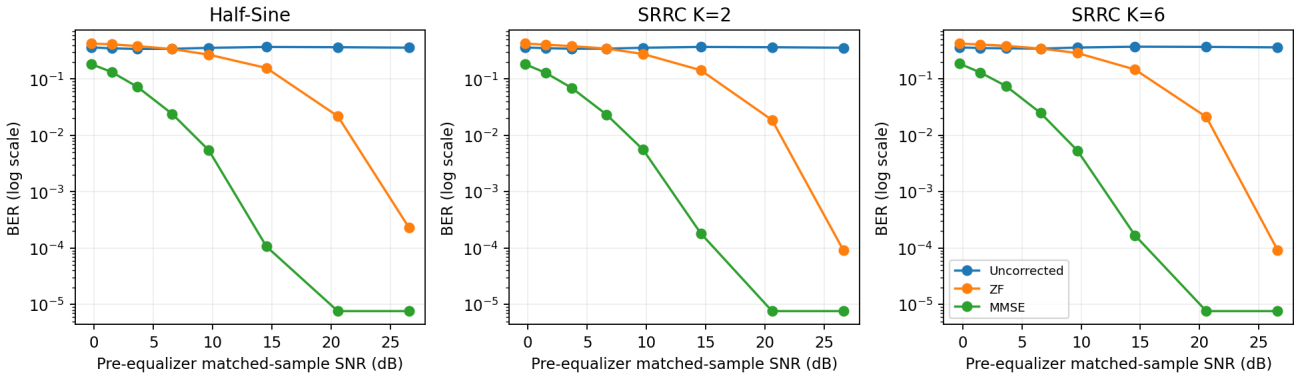


Figure 11. BER versus the pre-equalizer matched-decision SNR (not E_b/N_0).

7. BER, SNR, and image-quality measurements

The tabulated SNR is 10 log10(E{|y_noiseless[k]|²}/σ²), where y_noiseless is the matched-filter output sampled before equalization. Unit pulse energy makes σ² the AWGN decision-sample variance. The signal term includes multipath-induced ISI, so this definition is neither E_b/N_0 nor an equalizer-output SINR.

Pulse	σ	Pre-EQ SNR (dB)	Receiver	BER	Image PSNR (dB)
Half-Sine	0.10	20.56	ZF	0.02193	9.77
Half-Sine	0.10	20.56	MMSE	0.00000	39.09
Half-Sine	0.35	9.68	ZF	0.27046	6.16
Half-Sine	0.35	9.68	MMSE	0.00543	12.13
Half-Sine	0.50	6.58	ZF	0.34042	6.13
Half-Sine	0.50	6.58	MMSE	0.02411	8.28
SRRC K=6	0.10	20.57	ZF	0.02103	9.74
SRRC K=6	0.10	20.57	MMSE	0.00000	39.09
SRRC K=6	0.35	9.69	ZF	0.28571	6.12
SRRC K=6	0.35	9.69	MMSE	0.00534	13.19
SRRC K=6	0.50	6.59	ZF	0.34267	6.11
SRRC K=6	0.50	6.59	MMSE	0.02466	8.09

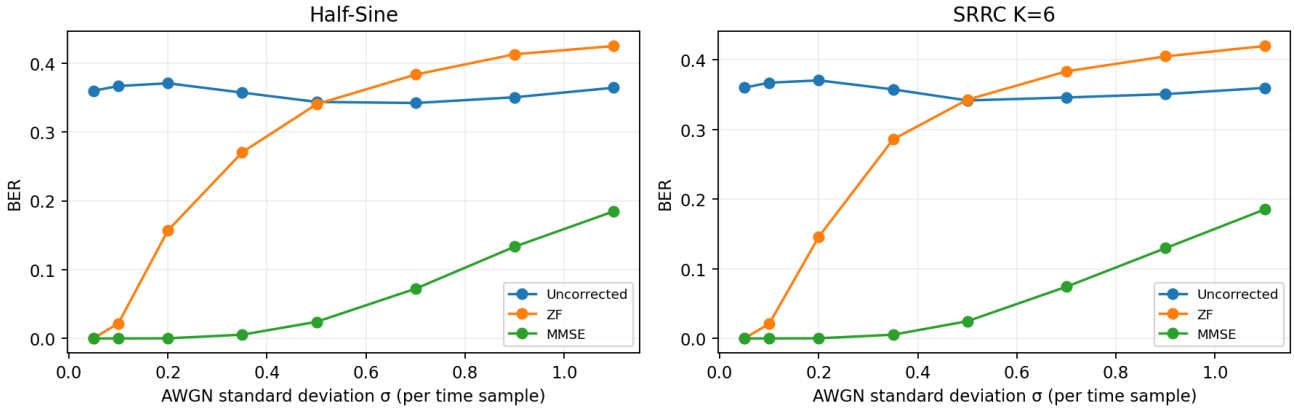


Figure 12. BER against the standard deviation of the noise injected before matched filtering.

At σ = 0.5, the Half-Sine receiver yields BER = 0.34042 (ZF) versus 0.02411 (MMSE); the K = 6 SRRC receiver yields BER = 0.34267 (ZF) versus 0.02466 (MMSE). These values are empirical for the specified channel, quantized payload, random seed, and approximate frequency-domain equalizer. They are not universal ZF/MMSE bounds.

8. Decision-point output SINR for ZF and MMSE

To characterize each equalizer separately, a noiseless equalizer output is projected onto the known bipolar transmitted symbols, yielding an empirical coherent gain g. For the same source bits and a separately generated noisy channel realization, all residual energy in z[k] − g a[k] is treated as error (additive noise, residual ISI, and finite-block effects). The measured output SINR is 10 log10(g² / mean{(z[k] − g a[k])²}). This estimator uses the actual filtered noise and does not rely on a white matched-noise approximation in the measured residual.

Pulse	σ	ZF output SINR (dB)	MMSE output SINR (dB)
Half-Sine	0.10	5.99	14.57
Half-Sine	0.35	-4.79	8.28
Half-Sine	0.50	-7.77	5.86
Half-Sine	0.90	-12.66	0.97
SRRC K=6	0.10	6.29	14.62
SRRC K=6	0.35	-4.42	8.27
SRRC K=6	0.50	-7.96	5.93
SRRC K=6	0.90	-12.74	0.98

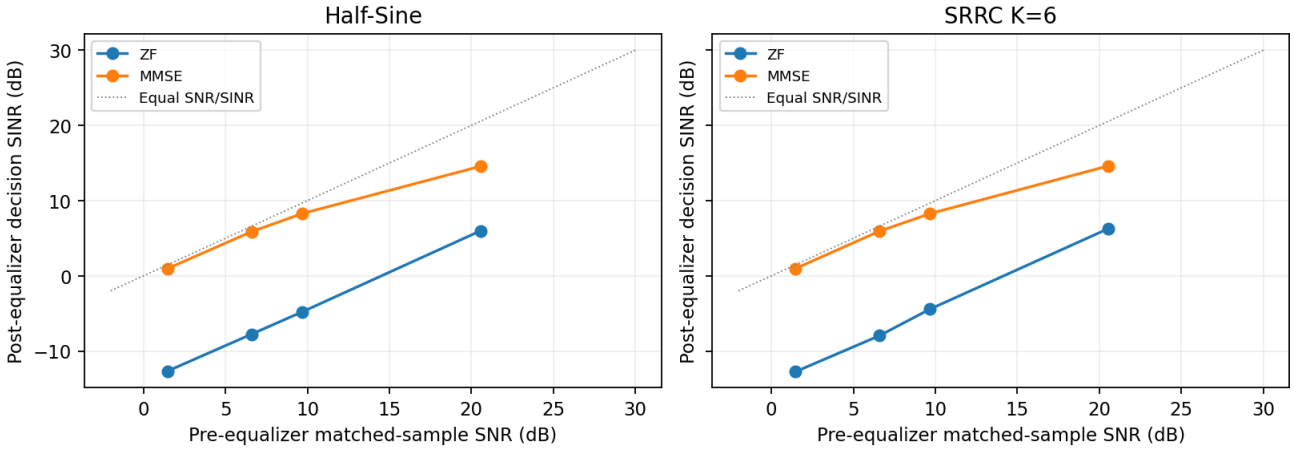


Figure 18. Empirical post-equalizer decision-point SINR contrasted with pre-equalizer matched-sample SNR.

At σ = 0.5 the Half-Sine ZF receiver has approximately −7.77 dB output SINR, while MMSE reaches 5.86 dB. The corresponding K = 6 SRRC results are −7.96 and 5.93 dB. This agrees with the measured BER ordering and demonstrates channel-inversion noise enhancement. Because the gain is estimated using the same known finite image bit sequence, these values are simulation-specific empirical SINR diagnostics rather than standardized E_b/N_0 measurements.

9. Reconstructed image fidelity

The receiver re-packs the hard decisions into bytes, reconstructs dequantized DCT coefficients, and applies inverse block DCT. BER reports binary correctness; PSNR reports visual reconstruction fidelity relative to the original image. Even a small BER can produce concentrated block artifacts when an erroneous bit changes a high-significance DCT coefficient.

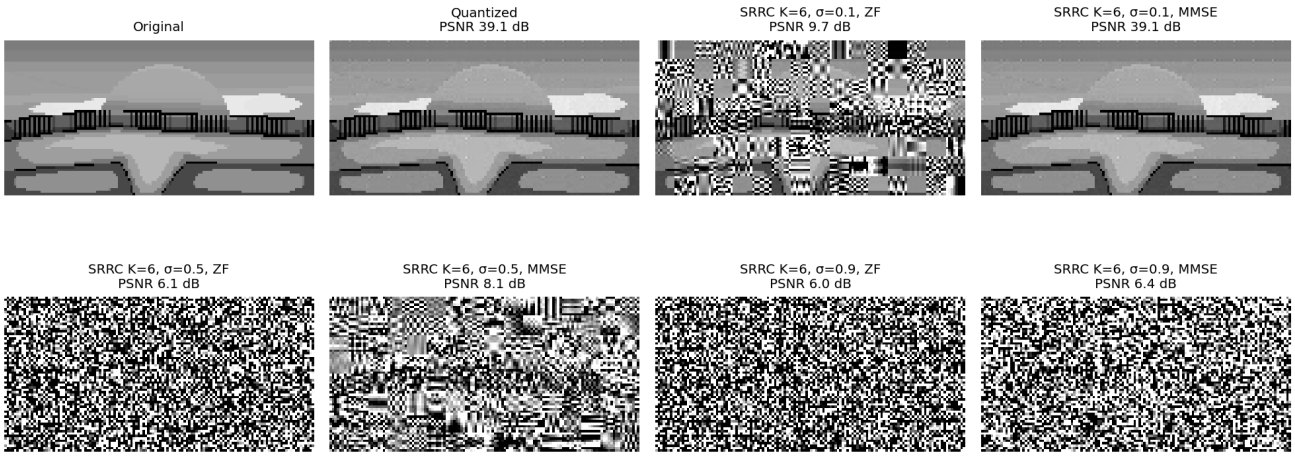


Figure 13. Image reconstructions for K = 6 SRRC at three noise levels with ZF and MMSE receivers.

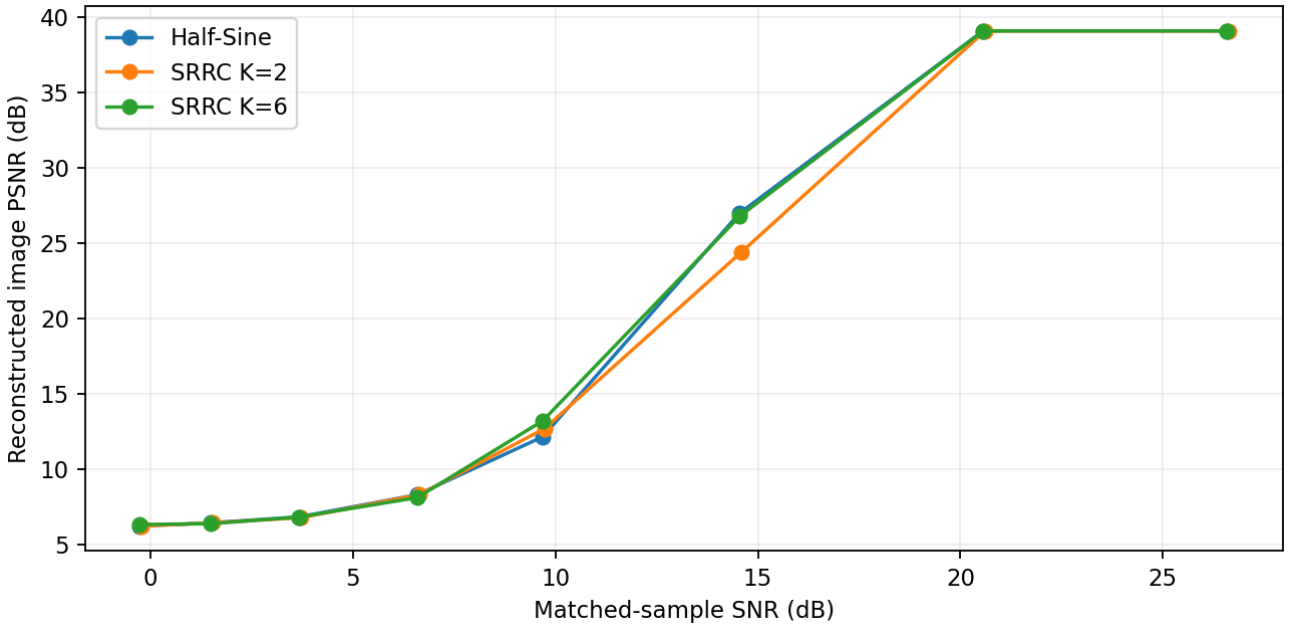


Figure 14. Matched-sample SNR versus reconstructed-image PSNR for MMSE equalization.

10. Three-image comparison

Three grayscale source images are tested under the same image dimensions (128 × 64), signal energy, noise level σ = 0.35, and primary channel. The original image is accompanied by resized Camera and Coins sample images from scikit-image. BER differences across source images reflect the distinct transmitted bit sequences, while PSNR additionally depends on the coefficient distribution and resulting image sensitivity.

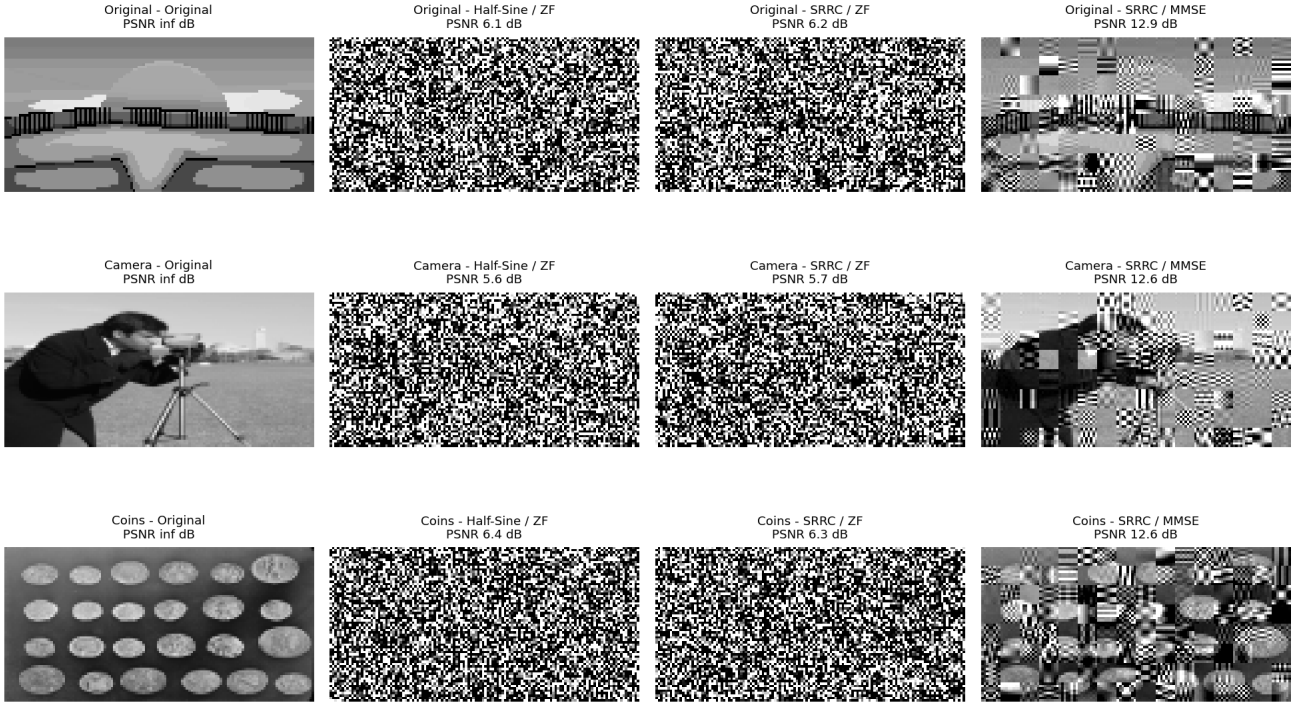


Figure 15. Original, ZF, and MMSE reconstructions of three images at σ = 0.35.

Image	Pulse	MMSE BER	MMSE PSNR (dB)
Original	Half-Sine	0.00476	13.27
Original	SRRC K=6	0.00479	12.91
Camera	Half-Sine	0.00433	14.20
Camera	SRRC K=6	0.00446	12.62
Coins	Half-Sine	0.00520	13.26
Coins	SRRC K=6	0.00621	12.64

11. SRRC pulse-span trade-offs

The SRRC roll-off is held at β = 0.5 while the finite pulse half-span K varies. Longer spans reduce the ideal-channel residual adjacent-symbol autocorrelation, but increase FIR support, filter group delay, and computation. The 99%-energy frequency measure is a finite-length FFT diagnostic rather than the ideal asymptotic SRRC bandwidth.

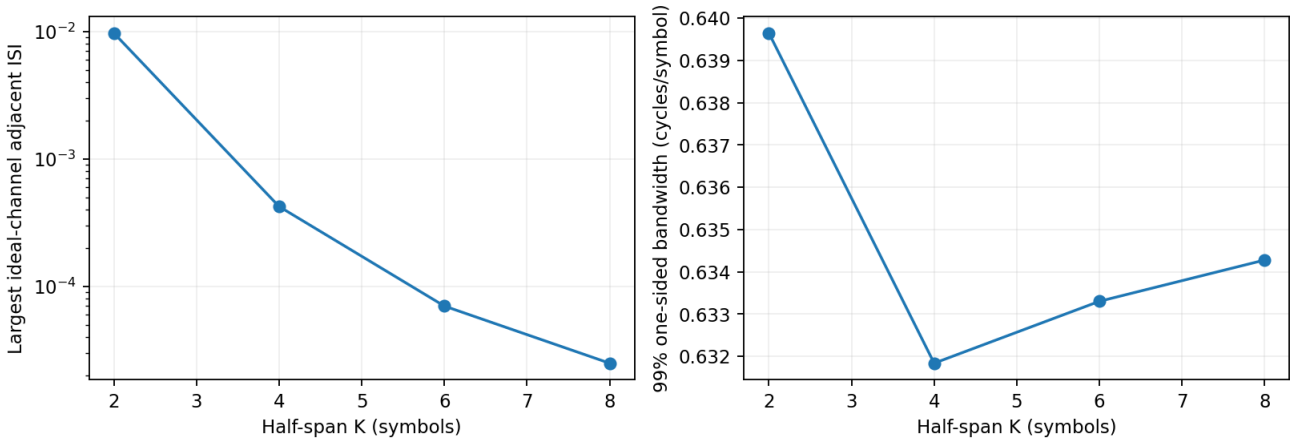


Figure 16. Pulse span versus ideal-channel adjacent-symbol ISI and 99% one-sided occupied bandwidth.

K	Pulse taps	Ideal adjacent ISI	99% BW (cycles/symbol)
2	129	0.009718	0.6396
4	257	0.000423	0.6318
6	385	0.000070	0.6333
8	513	0.000025	0.6343

The numeric comparison supports a qualified conclusion: both pulses can operate at equal energy and recover information, but bandwidth containment, truncation-induced residual ISI, channel response, and receiver regularization determine their observed BER. The Half-Sine pulse has the shortest support here; SRRC offers controlled roll-off and longer filtering delay. No pulse shape is universally optimal across noise/channel configurations.

12. Reproducibility

The accompanying Notebook saves all plots and CSV measurements. Its random generators use explicit seeds, and all physical signal-path convolutions use linear FFT convolution. The analysis assumes perfect channel and decision timing knowledge, AWGN, antipodal signaling, fixed packet dimensions, metadata delivered error-free, and circular frequency-domain equalization at the symbol block. These assumptions should be considered when extending the model to a practical radio link.

13. Alternative channels and robustness

The task-specific channel extension compares the original four-path response with two six-path symbol-spaced responses. Receiver parameters are recomputed from each effective channel rather than reused from the primary channel. Channel frequency dips, relative path gains, and regularization jointly explain why the same equalizer and noise standard deviation yield different error rates.

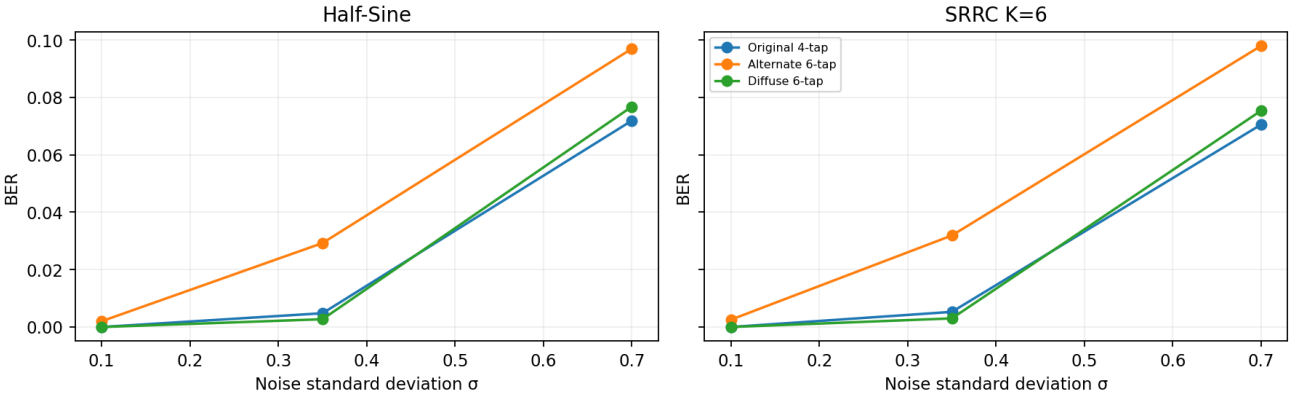


Figure 17. MMSE BER versus noise standard deviation for three channels and both pulse families.

Channel	Pulse	MMSE BER	Image PSNR (dB)
Original 4-tap	Half-Sine	0.00481	12.94
Original 4-tap	SRRC K=6	0.00528	12.62
Alternate 6-tap	Half-Sine	0.02925	10.85
Alternate 6-tap	SRRC K=6	0.03189	10.69
Diffuse 6-tap	Half-Sine	0.00272	15.81
Diffuse 6-tap	SRRC K=6	0.00301	14.88

Discussion and limitations

The results are controlled numerical experiments, not over-the-air measurements. The current model excludes clock and carrier recovery, channel estimation errors, coding, packet protocols, image-header transmission, nonstationary fading, and realistic colored front-end noise. SRRC matched-filter noise samples can be mildly correlated because pulses have finite span; the MMSE denominator approximates their covariance as diagonal. A finite FFT equalizer also uses a circular convolution model, which can create boundary artifacts.

Conclusion

The complete signal chain connects image-domain losses with physical-layer decisions. DCT quantization sets a nonzero reconstruction error floor even on a perfect link. Pulse shaping governs bandwidth occupancy and pulse overlap; multipath produces ISI, matched filtering controls AWGN detection sensitivity, and a noise-aware MMSE equalizer can avoid the severe noise enhancement of spectral inversion. Receiver design must be evaluated jointly using bit-error and image-quality metrics, especially as noise level, pulse span, and channel impulse response change.